

Vutukuri Sathvik

BACKEND ARCHITECT · DISTRIBUTED SYSTEMS · AI AGENTIC WORKFLOWS

11+ years as a **full-stack engineer** building complete products end to end — from large-scale **backend and distributed systems in Java** (high-throughput messaging, distributed scheduling, decisioning infrastructure) to **React** web portals. Deepest on the systems side, owning architecture through **production delivery at scale**, including platforms serving **1B+ requests/day**. Heavily leverage **AI-driven development** with Gemini and Claude for AI-assisted coding, automated unit/integration testing, and building **AI agentic workflows** into production microservices.

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EXPERIENCE

Member of Technical Staff I / Backend Architect

PayPal

Bengaluru, IN
Apr 2022 – Present

KAIROS — Communication Decision Platform

- Designed core components of **KAIROS**, PayPal's Communication Decision Platform that governs what every consumer is communicated, when, and on which channel — designed, delivered, and running in production.
- Architecting **AI-driven communication agents and production workflows** using Claude and Gemini APIs, with **function calling, tool integration and prompt orchestration** to automate intent evaluation, message eligibility scoring, and contextual payload generation.
- Pioneered **AI-driven engineering practices** across the team, extensively leveraging Gemini and Claude for automated code generation, complex refactoring, and comprehensive JUnit/Mockito test suites with **90%+ edge-case coverage**.
- Designed the **Intent Management System (IMS)** — the ingestion and management layer for decision intents flowing into the platform from producing systems.
- Architected the **Deferred-Intent Scheduler** — a durable, leaderless distributed timer on **Google Bigtable + Pulsar** sized for **25–30M deferred intents/day** with bursts up to **2M due at once**, using atomic **checkAndMutate** claims.
- Owned the Scheduler's end-to-end low-level design: hash-prefixed row-key schema, scan-claim-dispatch pipeline, cell-timestamp-based garbage collection, and failure recovery.

Server-Driven UI (SDUI) & SPF Architecture

- Led backend for PayPal's in-house **Server-Driven UI platform**, built from scratch and shipped to production, serving **1B+ requests/day** and powering core PayPal app & web experiences.
- Spearheaded planning and design of complex SDUI integrations using the **Server Presentation Framework (SPF)** to deliver cross-platform UI/UX across **iOS, Android & Web** entirely through server-side JSON configurations.
- Architected **flow inheritance models** within SPF to orchestrate dynamic user journeys, using **decision frames** to route users to contextually appropriate content frames.
- Enabled rapid feature iteration and **zero-release UI updates** by shifting UI logic to the backend, bypassing traditional App Store and Google Play release cycles.
- Integrated **Content Localization, ELMO A/B testing treatments, and Personalization Metadata** including AI-driven signals to deliver targeted and experimental user experiences.
- Guided cross-functional teams in adopting SDUI principles through the **SPF Admin UI Builder**, while enforcing component-based architecture for brand consistency and multi-tenant reusability.
- Designed integrations between backend domain services (**MFS layer**) and SPF server SDKs (**Java/Node**) to inject real-time context data, flow parameters, and personalized content into SPF templates before client rendering.
- Architected core SDUI services including **version management, page inheritance, schema versioning & database design**, enabling consistent UI delivery across iOS, Android & Web.

PayPal Communications — Comms Platform & AI Agents

- Architected and owned the end-to-end backend platform powering PayPal communications — **scheduling and delivery** across Push, Notification Center, Critical Alert & Silent Push for PayPal, Venmo & Xoom tenants.
- Driving the **communications platform revamp**, redesigning service architecture for scalability, multi-region support, and integration with the centralized decisioning layer.
- Built a production **AI agent for the Communications API**, integrating platform tools and APIs to enable natural-language interaction and automate communication workflows.
- Built and integrated an **MCP server with Claude**, exposing Communications platform capabilities as AI tools so teams can execute platform workflows through **agentic, natural-language interactions**.
- Designed and built **Beacon** — the production web portal used company-wide to schedule and manage communications, owning it end to end with **React**.

Verse — Evolution of SDUI / Content Architecture

- Extended the SDUI architecture into **Verse**, a next-generation content delivery platform for native iOS and Android clients, exposing page content through REST APIs using **pageld, locale and version-aware freshness** semantics.
- Evolved the backend content-delivery model with **in-memory caching** to minimize ECM reads, while enabling clients to retain cached content and refresh only when backend content versions change.
- Designed a reusable **content model aligned with Figma-driven UI templates**, enabling backend-managed composition and consistent rendering across native clients.
- Explored integration with **ELMO experimentation** to support A/B-tested experiences and dynamic content selection within the SDUI architecture.

PayPal SPOT Awards — Q3 2022 & Q4 2023

Senior Software Engineer

Informatica

Bengaluru, IN
Mar 2020 – Apr 2022

- Architected and built **SecureStats** end-to-end — security-flaw management platform with full backend, frontend, database design & APIs.
- Co-designed **Flaw Management Service & Notification Service** in a microservices architecture.
- Mentored 5 engineers on service design & architecture decisions. **INFASTAR Award Q1 2021**.

Senior Software Engineer

Envestnet Yodlee

Bengaluru, IN
Nov 2015 – Mar 2020

- Architected **Juggernaut** — microservice platform for automated issue resolution on Yodlee PFM. Owned end-to-end backend & frontend design.
- Designed observability stack with Spring Boot Actuator, Micrometer, Prometheus & Grafana; microservice infra with Eureka, Zuul & Hystrix.
- Yodlee Dream Team Award** — Juggernaut R&D (Q2 2018, Q2 2019).

SKILLS

AI & AGENTIC

LLM APIs (Claude, Gemini)
AI-Assisted Coding
AI-Assisted Testing
AI Agent Architecture
Agentic Workflows
Prompt Engineering
Function Calling RAG

LANGUAGES

Java 8-25 JavaScript SQL

DISTRIBUTED SYSTEMS

System Design
Distributed Scheduling
Event-Driven
Consistency & Idempotency
High Throughput
Fault Tolerance

CONCURRENCY

Virtual Threads
Structured Concurrency
CompletableFuture
Multithreading

BACKEND

Spring Boot GraphQL REST
Microservices gRPC

MESSAGING & STREAMING

Kafka Pulsar

DATA & STORAGE

Bigtable Oracle MongoDB

CLOUD & INFRA

GCP GKE Kubernetes Docker
Amazon S3

FRONTEND

React JS Redux JavaScript
Tailwind HTML/CSS

OBSERVABILITY

Datadog Grafana Prometheus
Micrometer OpenTelemetry

FOUNDATIONS

DSA OOP Design Patterns

EDUCATION

B.Tech — CSE

JNTU Kakinada · GPA 7.9/10
2011 – 2015

AWARDS

PayPal SPOT Awards
SDUI · Q3 2022, Q4 2023
Informatica INFASTAR
SecureStats · Q1 2021
Yodlee Dream Team
Juggernaut R&D · Q2 2018, Q2 2019
Yodlee Spot Awards
Q2/Q3 2016 · Q1 2018 · Q3 2019